Characteristics of polynomial functions

1

a -8 **b** -1 **c** 8184 **d** $16\,376$

2

a $\frac{17}{6}$ **b** $-\frac{16}{3}$ **c** $\frac{1529}{384}$ **d** 4

3

a 20 **b** $\frac{131\,950}{6561}$ **c** -1932 **d** 42

4

a Fall **b** Fall

5 It is impossible. A polynomial function that has degree n has at most $n-1$ turning points.

6 A polynomial function that has degree n has a graph with at most $n-1$ turning points.

7

a $(0,5)$ **b** 2

8

a Yes **b** No

9 False

10

a No **b** No **c** No **d** Yes

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Graphs of polynomial functions

11

a

i $x = 3, -1$

ii $(-\infty, -1), (3, \infty)$

iii $(-1, 3)$

b

i $x = -3, -1, 1$

ii $(-\infty, -3), (-1, 1)$

iii $(-3, -1), (1, \infty)$

c

i $x = -4, -2, 2$

ii $(-4, -2), (2, \infty)$

iii $(-\infty, -4), (-2, 2)$

d

i $x = 0, 2$

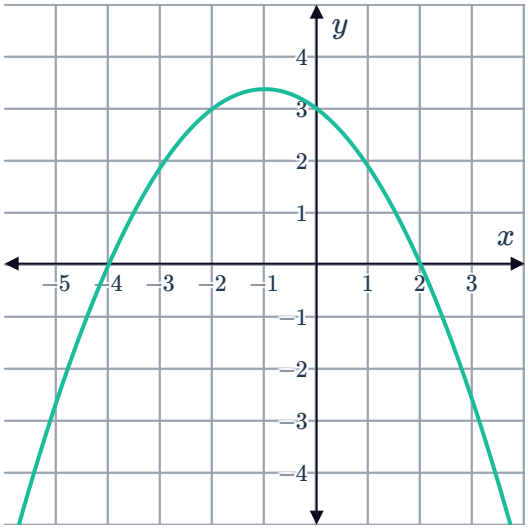
ii $(-\infty, 0), (2, \infty)$

iii $(0, 2)$

12

a 2

b



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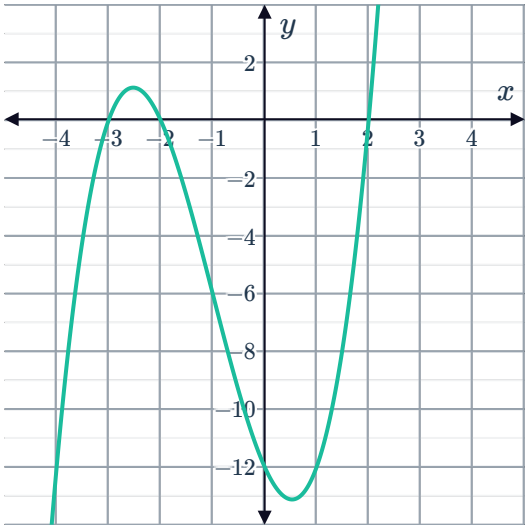
13

a

i $x = -3, -2, 2$

ii $y = -12$

iii

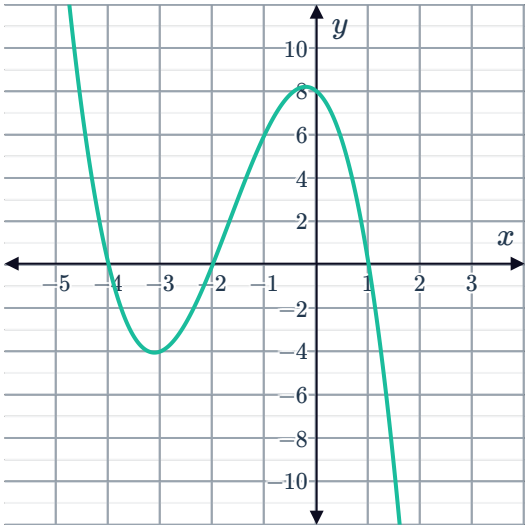


b

i $x = -4, -2, 1$

ii $y = 8$

iii

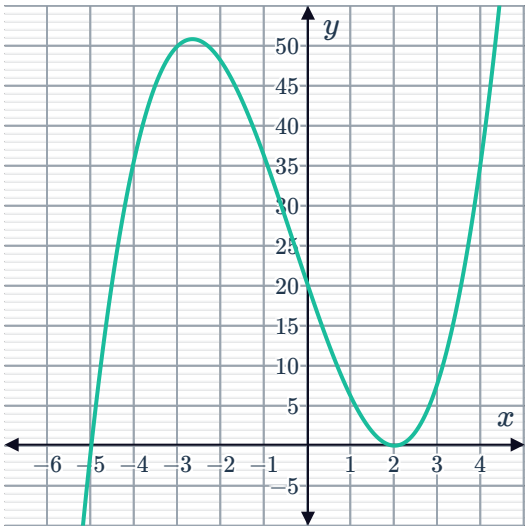


c

i $x = 2, -5$

ii $y = 20$

iii

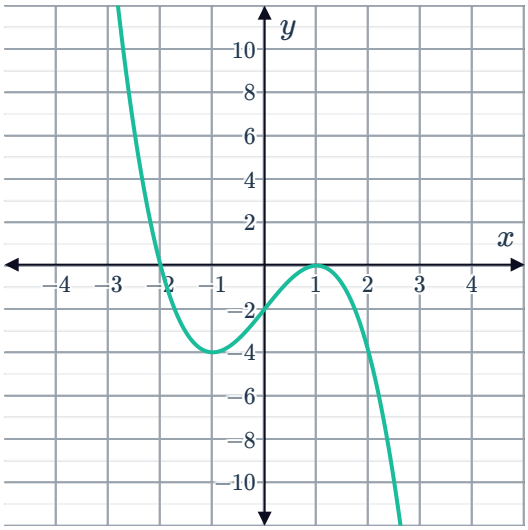


d

i $x = 1, -2$

ii $y = -2$

iii



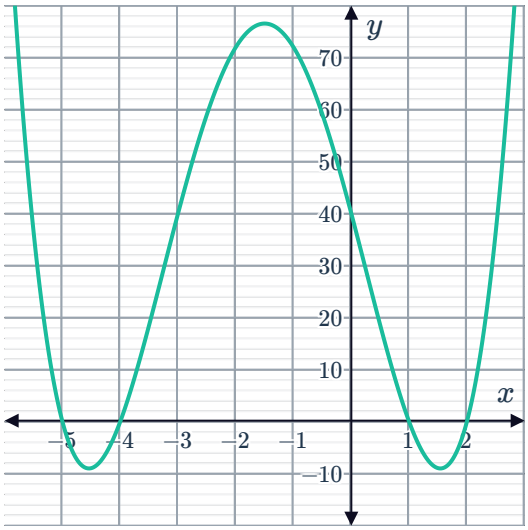
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e

i $x = 1, 2, -4, -5$

ii $y = 40$

iii

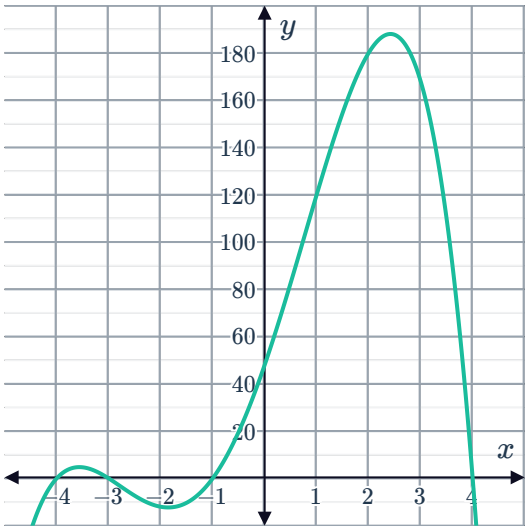


f

i $x = -1, -3, -4, 4$

ii $y = 48$

iii

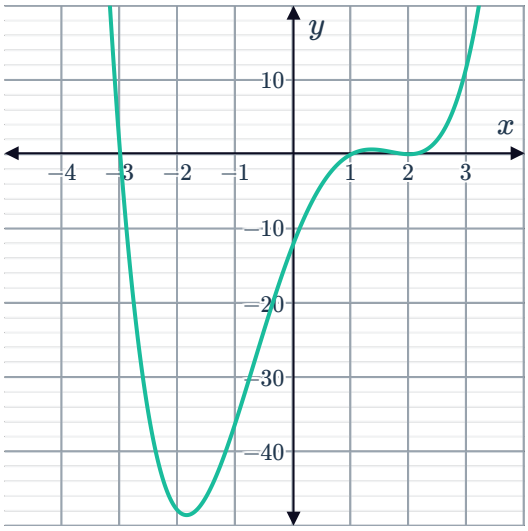


g

i $x = 2, -3, 1$

ii $y = -12$

iii

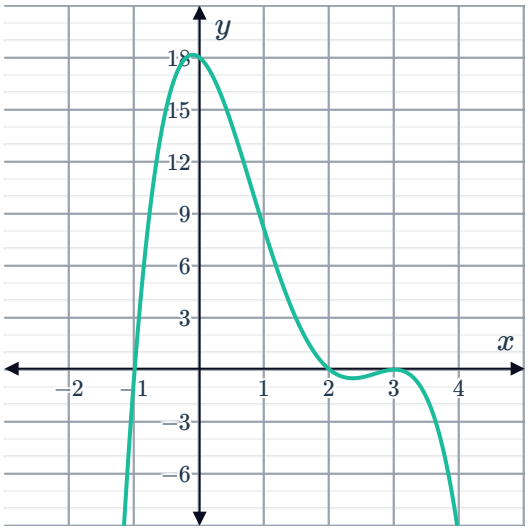


h

i $x = 3, -1, 2$

ii $y = 18$

iii



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i

i $x = 1, 4$

ii $y = 16$

iii

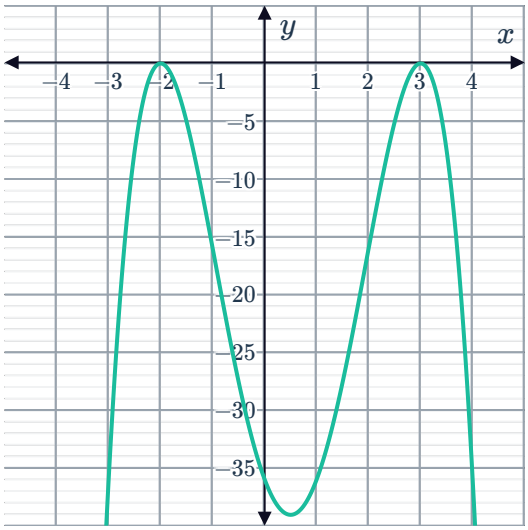


j

i $x = 3, -2$

ii $y = -36$

iii



14

a Rise

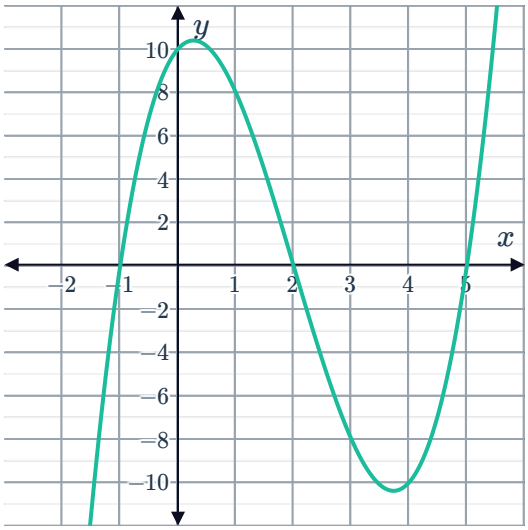
b Fall

c $y = (x - 2)(x - 5)(x + 1)$

d $x = 2, 5, -1$

e $y = 10$

f



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15

a 1

b Rise

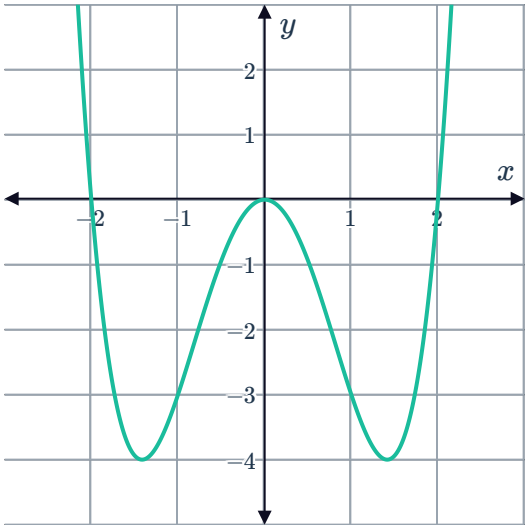
c Rise

d $y = x^2\,(x-2)\,(x+2)$

e $x = 0, -2, 2$

f $y = 0$

g



16

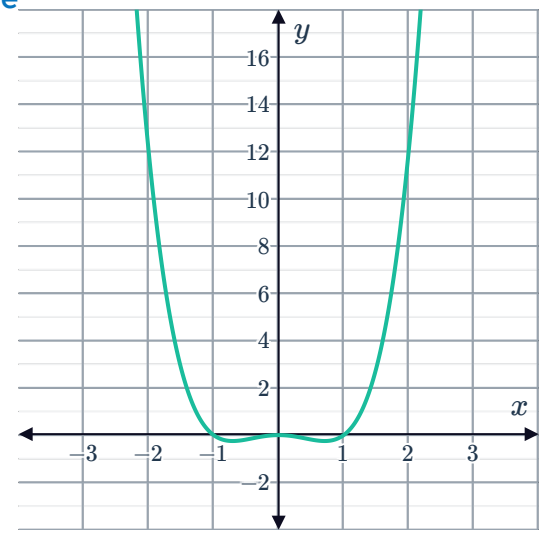
a 1

b Even

c Rise

d Rise

e



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17

a

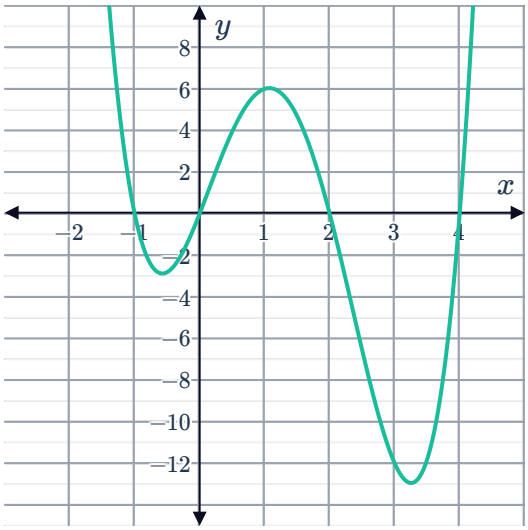
i $y \to \infty$

ii $y \to \infty$

iii $x = 0,\ -1,\ 2,\ 4$

iv $y = 0$

v



b

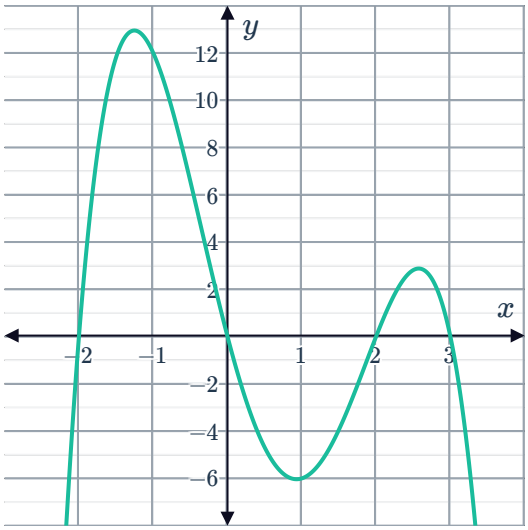
i $y \to -\infty$

ii $y \to -\infty$

iii $x = 0,\ -2,\ 2,\ 3$

iv $y = 0$

v



c

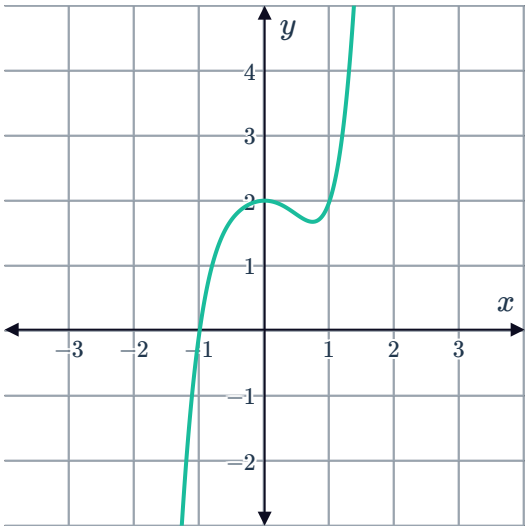
i $y \to -\infty$

ii $y \to \infty$

iii $x = -1$

iv $y = 2$

v



d

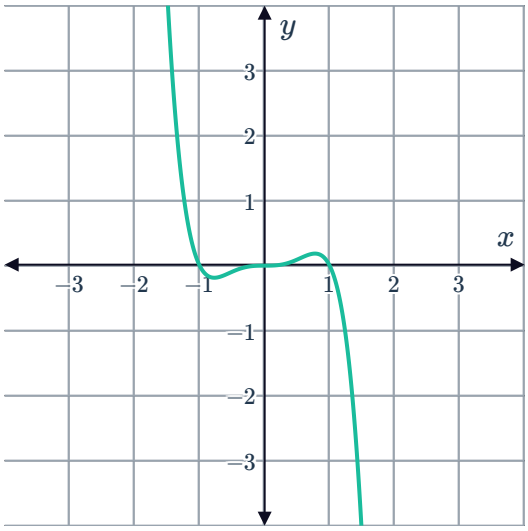
i $y \to \infty$

ii $y \to -\infty$

iii $x = 0,\ 1,\ -1$

iv $y = 0$

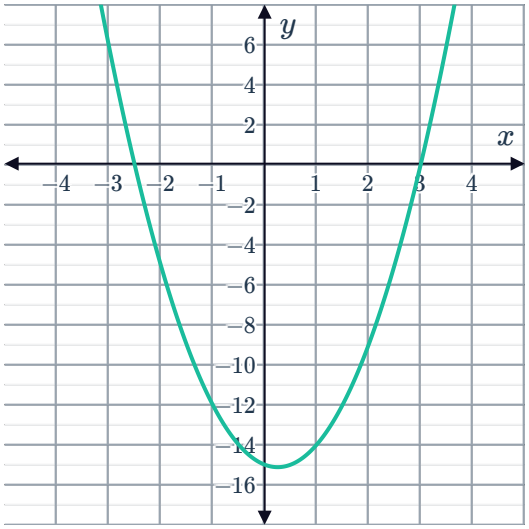
v



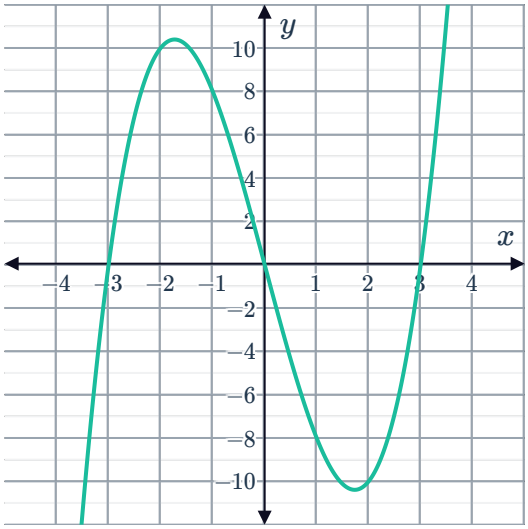
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18

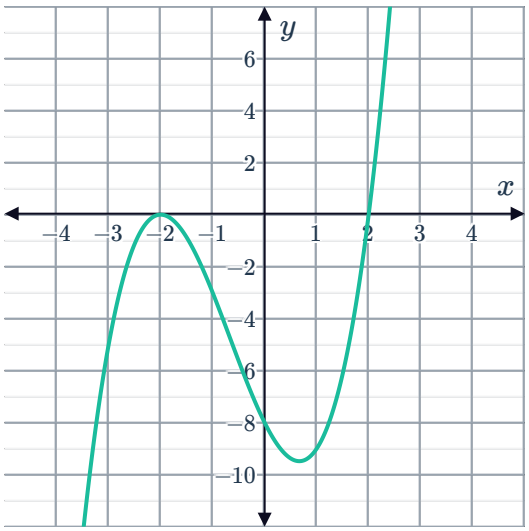
a



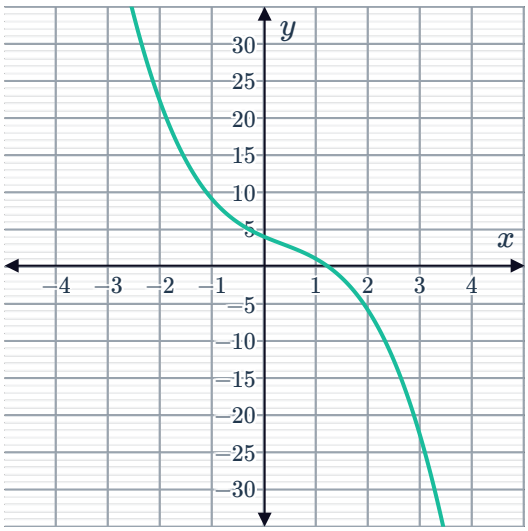
b



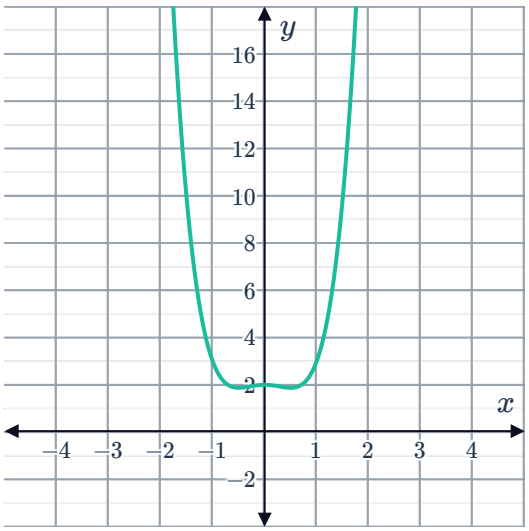
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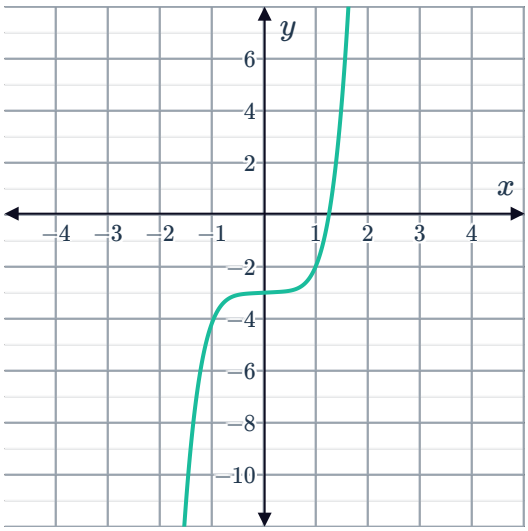
d



e

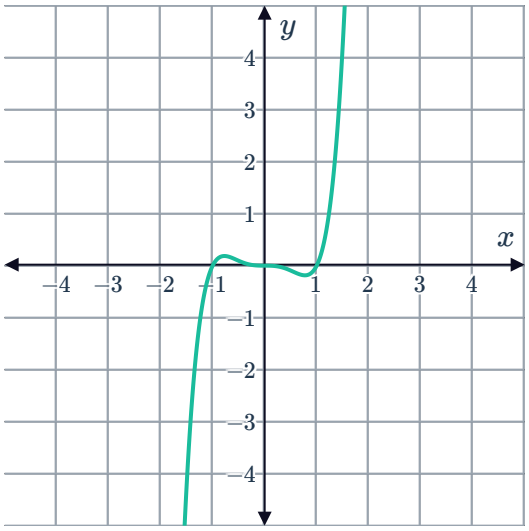


f

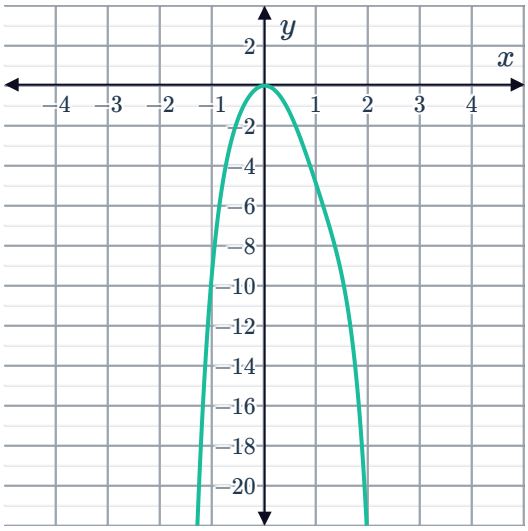


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g



h



Applications

19

a 54.24. At 23 years of age, a woman's level of good cholesterol should be around 54.24 mg/dL.

b 57.88. At 51 years of age, a woman's level of good cholesterol should be around 57.88 mg/dL.

20

a 16.658 L **b** 22.718 L **c** 25 years **d** 1995

21

a 9420 **b** 1081 **c** 56 years **d** 1986

22

a 390 m **b** 0 m

23

a 180 ft **b** 7 seconds

24

a $P = x^3 + 2x^2 - 10x + 7$ **b** Profit

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25

a

r	1	2	4	5	7
V	$\frac{4\pi}{3}$	$\frac{32\pi}{3}$	$\frac{128\pi}{3}$	$\frac{500\pi}{3}$	$\frac{1372\pi}{3}$

b **i** No **ii** Yes **iii** No **iv** No

c $r = 6$ m

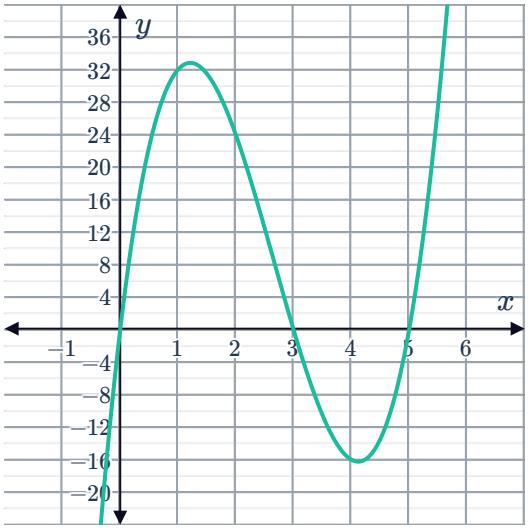
26

a 8 kg **b** 44.48 cm **c** 21 containers

27

a $V = 4x(3-x)(5-x)$ **b** $0 < x < 3$

c



d 24 ft^3

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